



**FINAL EXAMINATION
SEMESTER I, SESSION 2021/2022**

COURSE CODE	:	IFC 1024
COURSE NAME	:	CHEMISTRY
PROGRAMME	:	BRIDGING UTMSPACE
DURATION	:	3 HOURS
DATE	:	8 FEBRUARY 2022

INSTRUCTION TO CANDIDATES:

- 1.DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DOSO.
2. ANSWER ALL QUESTIONS.

WARNING!

Students caught copying/cheating during the examination will be liable for disciplinary actions and UTMSPACE may recommend the student to be expelled from the study.

QUESTION 1 (10 MARKS) $\text{mol of Mg} = \frac{25}{24} = 1.042 \text{ mol}$ $V = 0.5 \text{ L}$

1. 25 g of magnesium (Mg) is put into 500 mL of water. The reaction between the two produces magnesium hydroxide ($\text{Mg}(\text{OH})_2$) and hydrogen gas, and increases the temperature by 19°C .

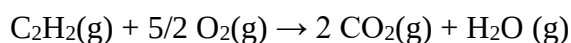
$$q = (525)(4.18)(19) \quad \Delta H = \frac{-41.7}{1.042} \quad 2\text{Mg} + 4\text{H}_2\text{O} \rightarrow 2\text{Mg}(\text{OH})_2$$

$$= 41.7 \text{ kJ} \quad = -40.02 \text{ kJ/mol}$$

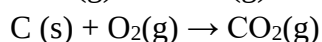
- (a) Calculate the enthalpy change, ΔH , of the reaction.
($C_s = 4.18 \text{ J/}^\circ\text{C}\cdot\text{g}$; $d = 1 \text{ g/mL}$)
- (b) Is this reaction endothermic or exothermic? *exo*
- (c) Based on your calculation, determine what is considered as the surrounding in this experiment. *heat released to surrounding*

(4 + 1 + 1 marks)

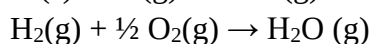
2. Given:



$$\Delta H = -1802 \text{ kJ/mol}$$

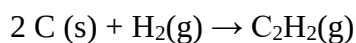


$$\Delta H = -703.5 \text{ kJ/mol}$$



$$\Delta H = -103.8 \text{ kJ/mol}$$

Calculate the enthalpy for the following reaction:



(4 marks)

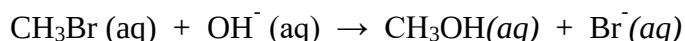
QUESTION 2 (10 MARKS)

$$r = k [\text{CH}_3\text{Br}]^2 [\text{OH}^-]^0$$

$$0.056 = k(0.05)^2(0.11)^0$$

$$k = 22.4 \text{ M}^{-2}\text{s}^{-1}$$

1. Consider the following reaction:



The rate law for this reaction is second order in CH_3Br and zero order in OH^- . When $[\text{CH}_3\text{Br}]$ is 0.050 M and $[\text{OH}^-]$ is 0.110 M, the reaction rate at 303 K is 0.056 M/s.

- Calculate the rate constant
- Determine the unit of the rate constant

(2 + 1 marks)

2. The gas phase decomposition of NO_2 , $2 \text{NO}_2(\text{g}) \rightarrow 2 \text{NO}(\text{g}) + \text{O}_2(\text{g})$, is studied at 383 °C, giving the following data:

Time (s)	$[\text{NO}_2]$ (M)
0.0	0.1000
5.0	0.0560 $k = 1.5714$
10.0	0.0314 $k = 2.1847$
15.0	0.0176 $k = 3.1212$
20.0	0.0098 $k = 4.6020$

- Determine the order of the reaction by using the mathematical method.
- Calculate the half life of the reaction.
- Do you expect the second half-life of the reaction to have the same value as the first one? Briefly explain your reasoning.

(a) $\therefore$ rate constants for first order are almost equal

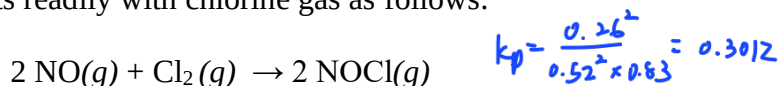
(4 + 1 + 2 marks)

(b) $t_{1/2} = \frac{0.693}{k} = 5.978 \text{ s}$

(c) No, the rate constant is different

QUESTION 3 (10 MARKS)

1. Nitric oxide (NO) reacts readily with chlorine gas as follows:



Given pressures of nitric oxide (NO) is 0.52 atm, chlorine (Cl₂) is 0.83 atm and NOCl is 0.26 atm.

- (a) Determine the equilibrium constant, K_p , of the reaction.
 (b) At 43 °C temperature, calculate the K_c of the reaction.
 (c) Determine the K_c for the reaction $4\text{NOCl}(g) \rightarrow 4\text{NO}(g) + 2\text{Cl}_2(g)$

(b) $K_p = K_c RT^{\Delta n}$ (c) $K'_c = (K_c)^{-2}$
 $0.3012 = K_c (0.0821 \times 316)^{-1}$, $K_c = 7.8143$ $= 0.0164$ (2 + 2 + 2 marks)

2. Iron (III) oxide, (Fe₂O₃), has a solubility of 2.3×10^{-3} g/10 mL of water. Determine the solubility-product constant, K_{sp} , of Fe₂O₃ in water.

Solubility = m.m. $\times$ molar solubility
 $2.3 \times 10^{-3} = 160 \times \text{molar solubility}$
 molar solubility = 1.4375×10^{-5}
 $K_{sp} = [\text{Fe}^{3+}]^2 [\text{O}^{2-}]^3$
 $= (2x)^2 (3x)^3$
 $= (2 \times 1.4375 \times 10^{-5})^2 (3 \times 1.4375 \times 10^{-5})^3$
 $= 3.683 \times 10^{-24} \text{ M}^5$ (4 marks)

QUESTION 4 (15 MARKS)

1. When salt becomes a solution, it can either be acidic or basic salt solutions. Sodium chloride, NaCl is a salt with numerous applications, including flavour enhancement, ice removal in winter, and serves as chlorine feedstock. On the other hand, sodium amide, NaNH₂ serves as a catalyst, a nucleophilic reagent and has important part in the industrial production of indigo dye.

- (a) The aqueous salt solution of NaCl is neutral. Explain this statement using the salt dissociation equation in water.
- (b) The aqueous salt solution of NaNH₂ is basic. Explain this statement using the salt dissociation equation in water.

(3+3 marks)

2. During Christmas, Santa Claus found a bottle of used vinegar in his kitchen. Its chemical formula is CH₃COOH, also known as acetic acid. He wants to make a Caesar salad for his reindeers. In order to do that, you are requested to help Santa Claus to find the pH of the acetic acid in the 0.200 M solution. Given K_a = 1.85 × 10⁻⁵.

$$\begin{aligned} \text{pH} &= -\log[\text{H}^+] \\ &= -\log(0.0019) \\ &= 2.72 \end{aligned}$$

	$\text{CH}_3\text{COOH} \rightleftharpoons \text{CH}_3\text{COO}^- + \text{H}^+$				
I	0.2	0	0		$1.85 \times 10^{-5} = \frac{x^2}{0.2}$
C	-x	+x	+x		
E	0.2-x	x	x		$x = 0.0019$

(5 marks)

3. Buffer solutions are used by the human body to maintain a constant pH for biochemical processes such as enzyme activity. In addition, buffer is used in baby shampoo and soap to maintain the pH, thereby preventing rashes on the baby. Given an ethanoic acid, CH₃COOH with K_a = 1.75 × 10⁻⁵. $K_b = 571.43 \times 10^{-12}$

- (a) Calculate the pH of a buffer solution containing 0.100 mol/L ethanoic acid and 0.100 mol/L sodium ethanoate using the Henderson-Hasselbalch equation.
- (b) Give the proportion (ratio) for [CH₃COO⁻] : [CH₃COOH] with the same concentration that we should mix to produce a buffer solution of pH 7.00.

$$\begin{aligned} \text{CH}_3\text{COONa} &\rightleftharpoons \text{CH}_3\text{COO}^- + \text{Na}^+ \\ \text{CH}_3\text{COO}^- + \text{H}_2\text{O} &\rightleftharpoons \text{CH}_3\text{COOH} + \text{OH}^- \end{aligned}$$

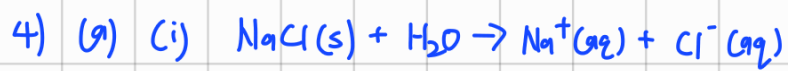
$$\begin{aligned} \text{pOH} &= -\log(571.43 \times 10^{-12}) + \log\left(\frac{0.1}{0.1}\right) \\ &= 9.24 \\ \text{pH} &= 4.76 \end{aligned}$$

(1+3 marks)

$$7 = -\log(571.43 \times 10^{-12}) + \log\left(\frac{[\text{c. base}]}{[\text{acid}]}\right)$$

$$\frac{[\text{c. base}]}{[\text{acid}]} = \frac{0.0575}{1}$$

$$[\text{c. base}] : [\text{acid}] = 0.0575 : 1$$



Na^+ ion is belongs to strong base NaOH

Cl^- ion is belongs to strong acid HCl

the reaction b/w strong acid and base will formed neutral salt



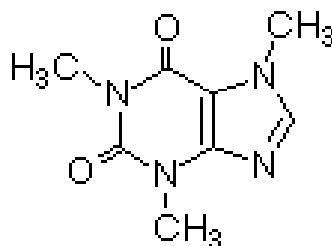
Na^+ ion is belongs to strong base NaOH

NH_2^- ion is belongs to weak acid NH_3

the reaction will produce a basic salt

QUESTION 5 (15 MARKS)

1. Caffeine is a purine alkaloid that occurs naturally in coffee and tea, and has a role as a central nervous system stimulant. Redraw the figure given below in your answer sheet and circle **FOUR** functional groups that you can find in the caffeine molecule.



Caffeine

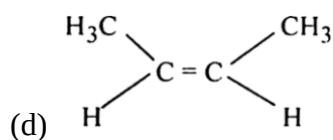
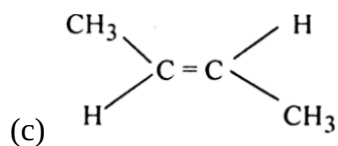
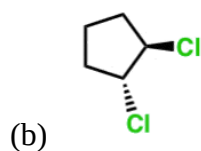
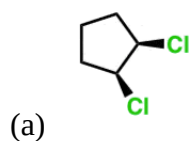
(4 marks)

2. Draw the structural formula for the following molecules :

- (a) 1,2-dimethyl cyclobutane
- (b) 2,2-dimethylpropane
- (c) 1,4-hexadiene
- (d) styrene
- (e) ortho-hydroxybenzoic acid
- (f) methylcyclopropane
- (g) 2-propanone

(7 marks)

3. Write the IUPAC name for the given molecules below :



(4 marks)



- 3) (a) cis-1,2-dichloropentane
 (b) trans-1,2-dichloropentane
 (c) trans-2-butene
 (d) cis-2-butene

1												18					
IA											VIIIA						
1	2											2					
H	He											He					
1.01												4.00					
3	4											10					
Li	Be											Ne					
6.94	9.01											20.18					
11	12											18					
Na	Mg											Ar					
22.99	24.31											39.95					
19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
39.1	40.08	44.96	47.88	50.94	52.00	54.94	55.85	58.93	58.69	63.55	65.39	69.72	72.61	74.92	78.96	79.90	83.80
37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
85.47	87.62	88.91	91.22	92.91	95.94	(98)	101.07	102.91	106.42	107.87	112.41	114.82	118.71	121.76	127.6	126.9	131.29
55	56	57	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86
Cs	Ba	La*	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
132.9	137.3	138.9	178.5	180.9	183.9	186.2	190.2	192.2	195.1	197.0	200.6	204.4	207.2	209	(209)	(210)	(222)
87	88	89	104	105	106	107	108	109	110	111							
Fr	Ra	Ac^	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg							
(223)	(226)	(227)	(261)	(262)	(263)	(264)	(265)	(268)	(271)	(272)							
</																	

LIST OF SELECTED CONSTANT VALUES

Speed of light in a vacuum	c	=	$3.0 \times 10^8 \text{ m s}^{-1}$
Avogadro's number	N_A	=	$6.02 \times 10^{23} \text{ mol}^{-1}$
Faraday constant	F	=	$9.65 \times 10^4 \text{ C mol}^{-1}$
Planck constant	h	=	$6.6256 \times 10^{-34} \text{ J s}$
Reduced Planck constant	$\hbar$	=	$1.054 \times 10^{-34} \text{ J s}$
Rydberg constant	R_H	=	$1.097 \times 10^7 \text{ m}^{-1}$ = $2.18 \times 10^{-14} \text{ J}$
Molar of gases constant	R	=	$8.314 \text{ J K}^{-1} \text{ mol}^{-1}$ = $0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1}$
Boltzmann constant	k	=	$1.3807 \times 10^{-27} \text{ J K}^{-1}$
Mass of proton	M_p	=	$1.672 \times 10^{-27} \text{ kg}$
Electronic Bohr magneton	m_B	=	$9.2741 \times 10^{-24} \text{ J T}^{-1}$
Nuclear Bohr magneton	b_N	=	$5.05 \times 10^{-27} \text{ J T}^{-1}$
Vapour pressure of water	P_{water}	=	23.8 torr
Electron charge	e^-	=	$1.602 \times 10^{-19} \text{ C}$

UNIT AND CONVERSION FACTOR

Energy	$1 \text{ J} = 1 \text{ kg m}^2 \text{ s}^{-2} = 1 \text{ N m} = 10^7 \text{ erg}$ $1 \text{ calorie} = 4.184 \text{ Joule}$ $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$ $1 \text{ amu} = 1.66 \times 10^{-27} \text{ kg}$
Pressure	$1 \text{ atm} = 760 \text{ mm Hg} = 760 \text{ torr} = 101.325 \text{ kPa} = 101325 \text{ N m}^{-2}$

FORMULAE

$$\frac{1}{\lambda_{\text{vac}}} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$4r = a\sqrt{2}$$

$$\text{rate} = \sqrt{\frac{1}{\text{density}}}$$

$$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{0.0592}{n} \log Q$$

STANDARD REDUCTION POTENTIAL ELECTROCHEMICAL SERIES

Half-Reaction	E (Volts)
$\text{K}^+ + \text{e}^- \rightleftharpoons \text{K}$	-2.924
$\text{Ba}^{2+} + 2\text{e}^- \rightleftharpoons \text{Ba}$	-2.900
$\text{Ca}^{2+} + 2\text{e}^- \rightleftharpoons \text{Ca}$	-2.760
$\text{Na}^+ + \text{e}^- \rightleftharpoons \text{Na}$	-2.712
$\text{Mg}^{2+} + 2\text{e}^- \rightleftharpoons \text{Mg}$	-2.375
$\text{H}_2 + 2\text{e}^- \rightleftharpoons 2\text{H}^+$	-2.230
$\text{Al}^{3+} + 3\text{e}^- \rightleftharpoons \text{Al}$	-1.706
$\text{Mn}^{2+} + 2\text{e}^- \rightleftharpoons \text{Mn}$	-1.040
$\text{Zn}^{2+} + 2\text{e}^- \rightleftharpoons \text{Zn}$	-0.763
$\text{Cr}^{3+} + 3\text{e}^- \rightleftharpoons \text{Cr}$	-0.740
$\text{S} + 2\text{e}^- \rightleftharpoons \text{S}^{2-}$	-0.508
$2\text{CO}_2 + 2\text{H}^+ + 2\text{e}^- \rightleftharpoons \text{H}_2\text{C}_2\text{O}_4$	-0.490
$\text{Cr}^{3+} + \text{e}^- \rightleftharpoons \text{Cr}^{2+}$	-0.410
$\text{Fe}^{2+} + 2\text{e}^- \rightleftharpoons \text{Fe}$	-0.409
$\text{Co}^{2+} + 2\text{e}^- \rightleftharpoons \text{Co}$	-0.280
$\text{Ni}^{2+} + 2\text{e}^- \rightleftharpoons \text{Ni}$	-0.230
$\text{Sn}^{2+} + 2\text{e}^- \rightleftharpoons \text{Sn}$	-0.136
$\text{Pb}^{2+} + 2\text{e}^- \rightleftharpoons \text{Pb}$	-0.126
$\text{Fe}^{3+} + 3\text{e}^- \rightleftharpoons \text{Fe}$	-0.036
$2\text{H}^+ + 2\text{e}^- \rightleftharpoons \text{H}_2$	0.000
$\text{S}_4\text{O}_6^{2-} + 2\text{e}^- \rightleftharpoons 2\text{S}_2\text{O}_3^{2-}$	0.089
$\text{Sn}^{4+} + 2\text{e}^- \rightleftharpoons \text{Sn}^{2+}$	0.150
$\text{Cu}^{2+} + \text{e}^- \rightleftharpoons \text{Cu}^+$	0.158
$\text{Cu}^{2+} + 2\text{e}^- \rightleftharpoons \text{Cu}$	0.340
$\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightleftharpoons 4\text{OH}^-$	0.401
$\text{Cu}^+ + \text{e}^- \rightleftharpoons \text{Cu}$	0.522
$\text{I}_3^- + 2\text{e}^- \rightleftharpoons 3\text{I}^-$	0.534
$\text{MnO}_4^- + 2\text{H}_2\text{O} + 3\text{e}^- \rightleftharpoons \text{MnO}_2 + 4\text{OH}^-$	0.588
$\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightleftharpoons \text{H}_2\text{O}_2$	0.682
$\text{Fe}^{3+} + \text{e}^- \rightleftharpoons \text{Fe}^{2+}$	0.770
$\text{Hg}_2^{2+} + 2\text{e}^- \rightleftharpoons \text{Hg}$	0.796
$\text{Cl}_2 + 2\text{e}^- \rightleftharpoons 2\text{Cl}^-$	1.360

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